

Generation and Characterization of Attosecond Pulses from an X-ray Free-electron Laser

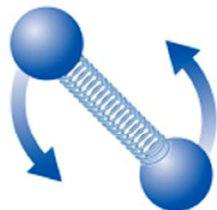
Siqi Li

Sep 2019



10 ps

Molecular rotation ~ ps



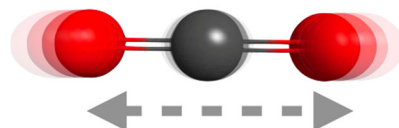
1 ps

LCLS Day 1

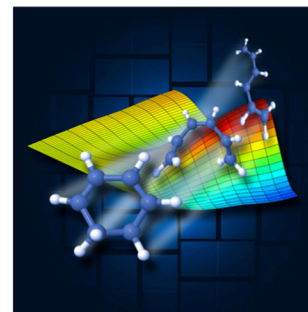
100 fs

LCLS 2018

10 fs



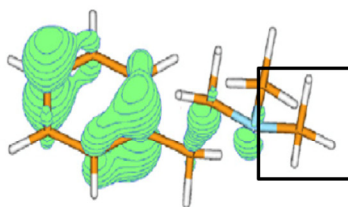
Molecular vibration ~10s fs



Molecular rearrangement ~100s fs

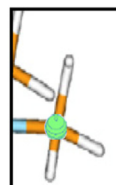
1 fs

Our goal



Valence electron motion ~ 100s as
J. Chem Phys. 8 129(10):104305 (2008)

0.1 fs

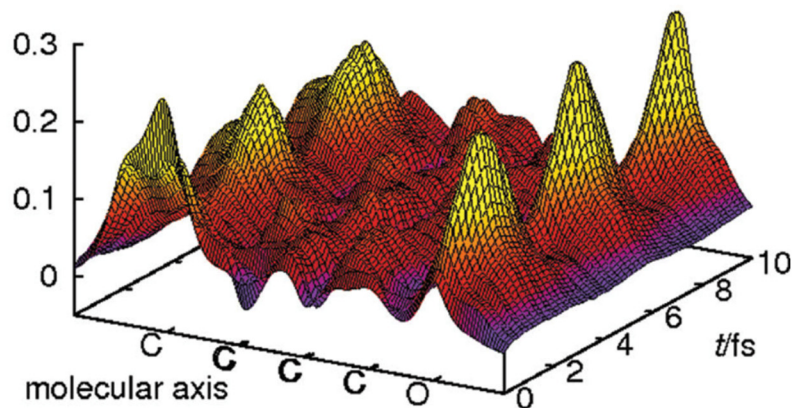


Inner electron motion ~ 10 as

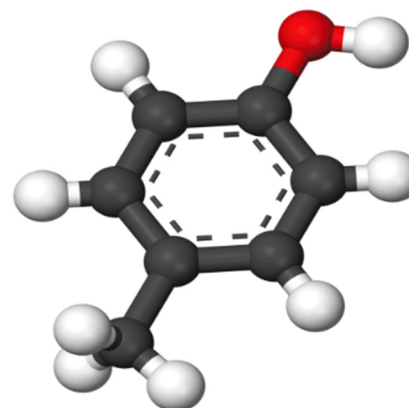
Why sub-femtosecond x-rays?

- Probe length scale of angstrom.
- Probe electron motion on their natural time scale: 100 as~ a few fs.

hole density $Q(z,t)$



4-Methylphenol

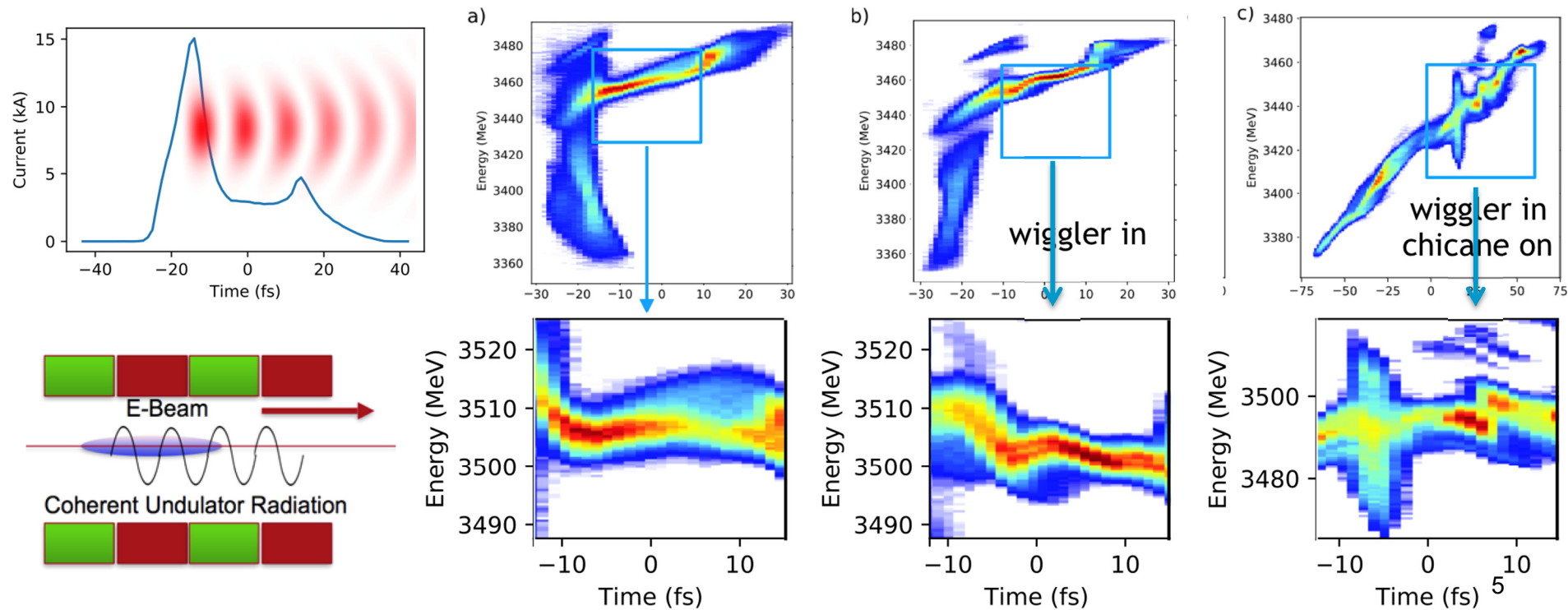


Outline

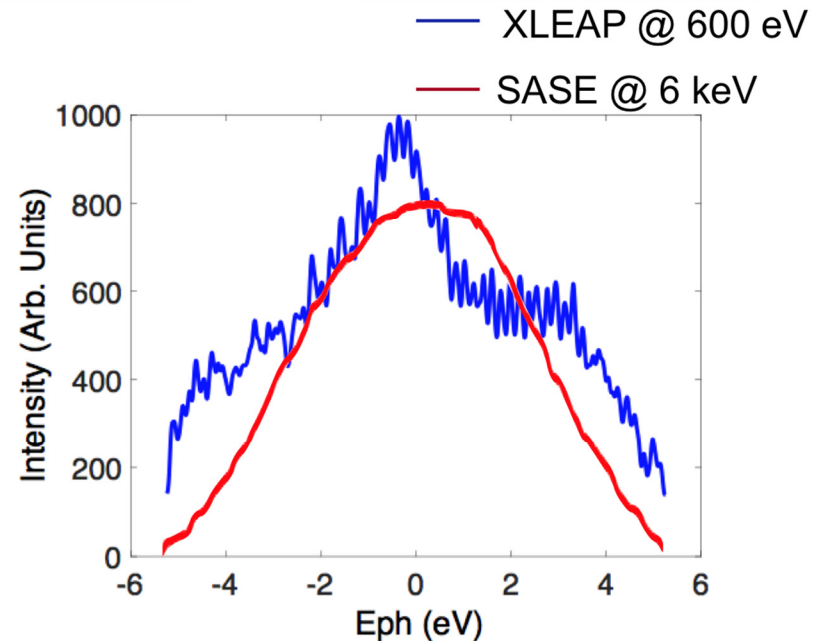
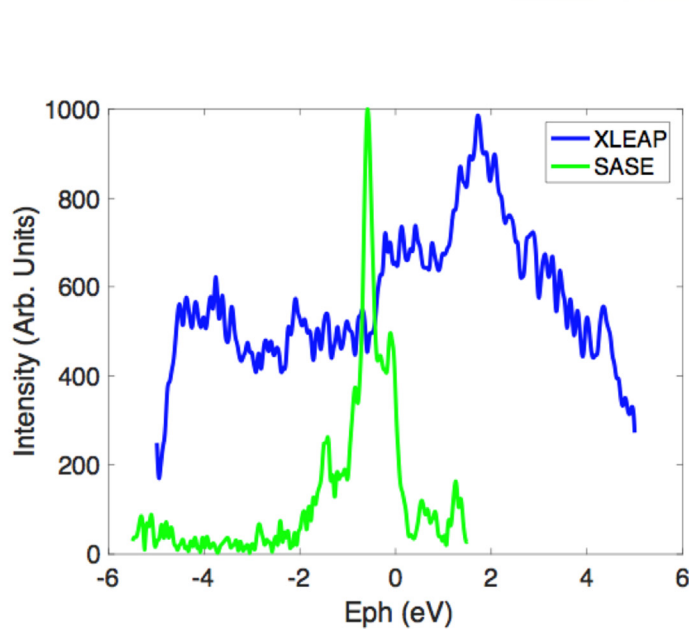
- Introduction: why sub-femtosecond x-rays?
- **XLEAP**: **X**-ray **L**aser **E**nhanced **A**ttosecond **P**ulses
- Angular streaking
- Data analysis
- Applications
- Conclusions

XLEAP: X-ray Laser Enhanced Attosecond Pulses

SLAC



XLEAP: X-ray Laser Enhanced Attosecond Pulses

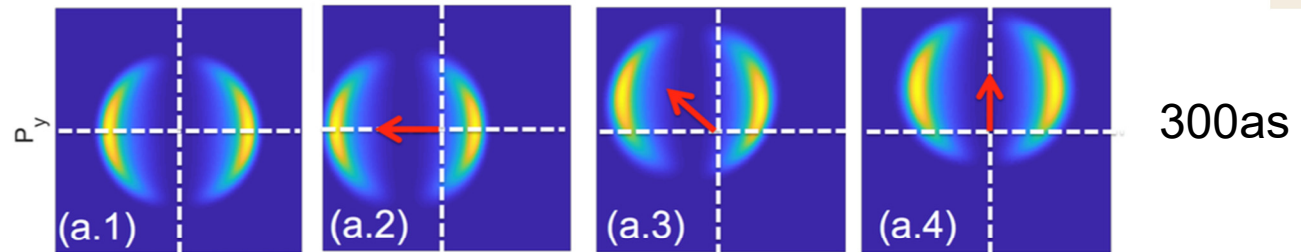


- 6 times broader than single-spike SASE at similar energy
- Broader than hard x-ray SASE too

Streaking simulations

SLAC

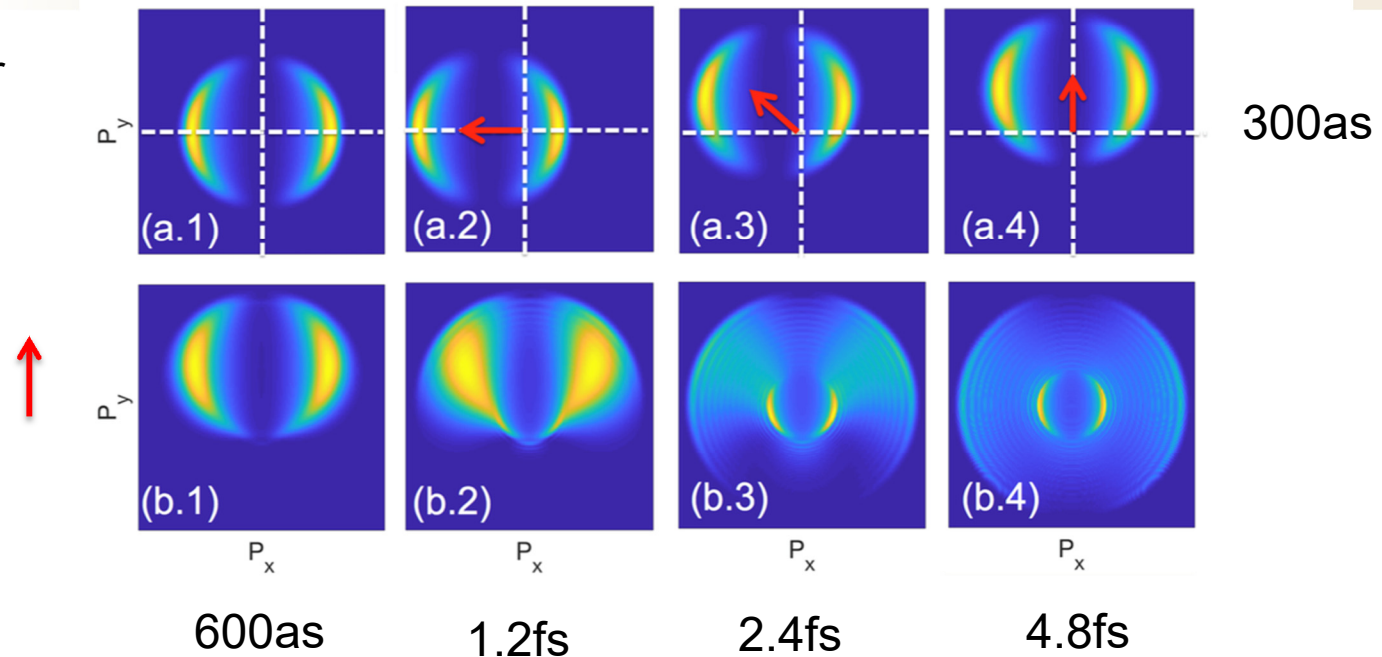
1.3 μ m streaking laser
~4.3fs period



Streaking simulations

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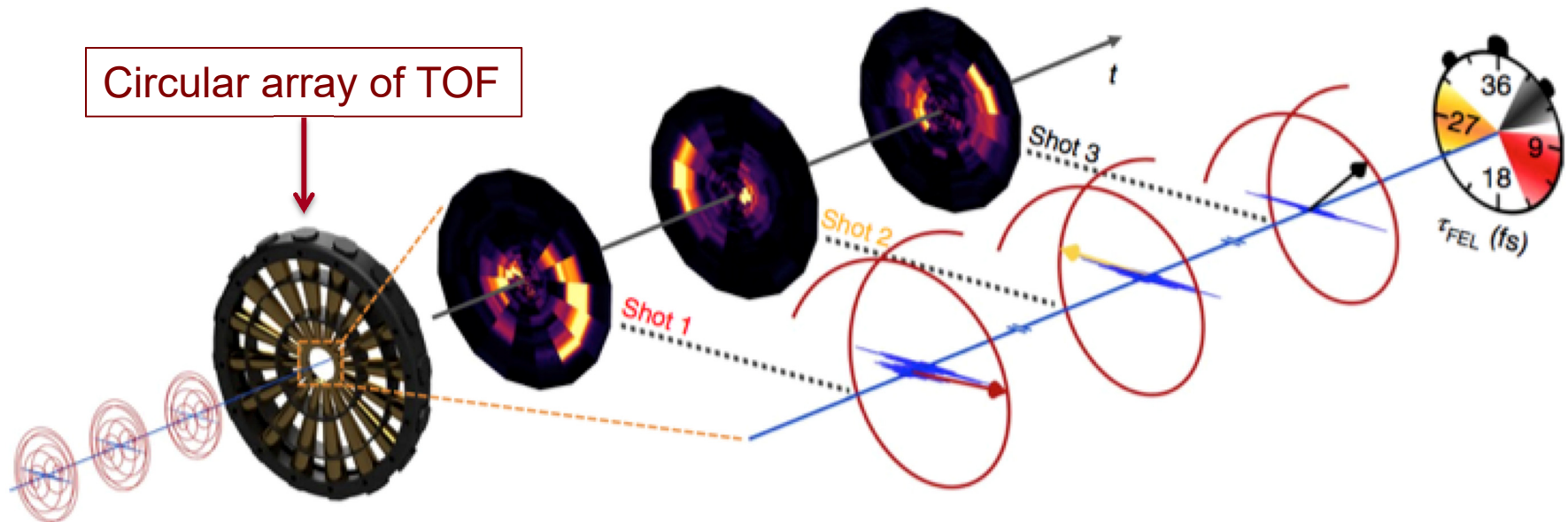
1.3um streaking laser
~4.3fs period



$$\tau < T_L$$

Recent demonstration with fs pulses

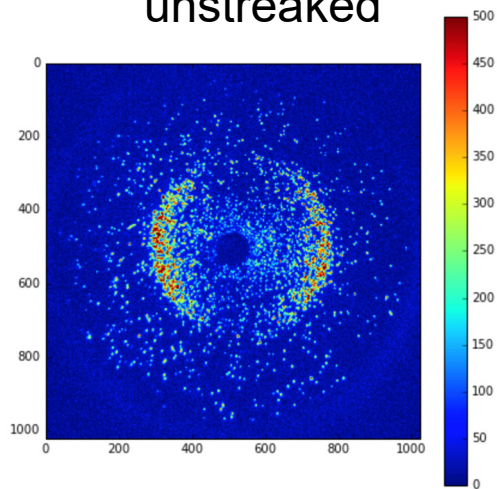
Hartmann, N., et al. "Attosecond time–energy structure of X-ray free-electron laser pulses." *Nature Photonics* 12.4 (2018): 215.



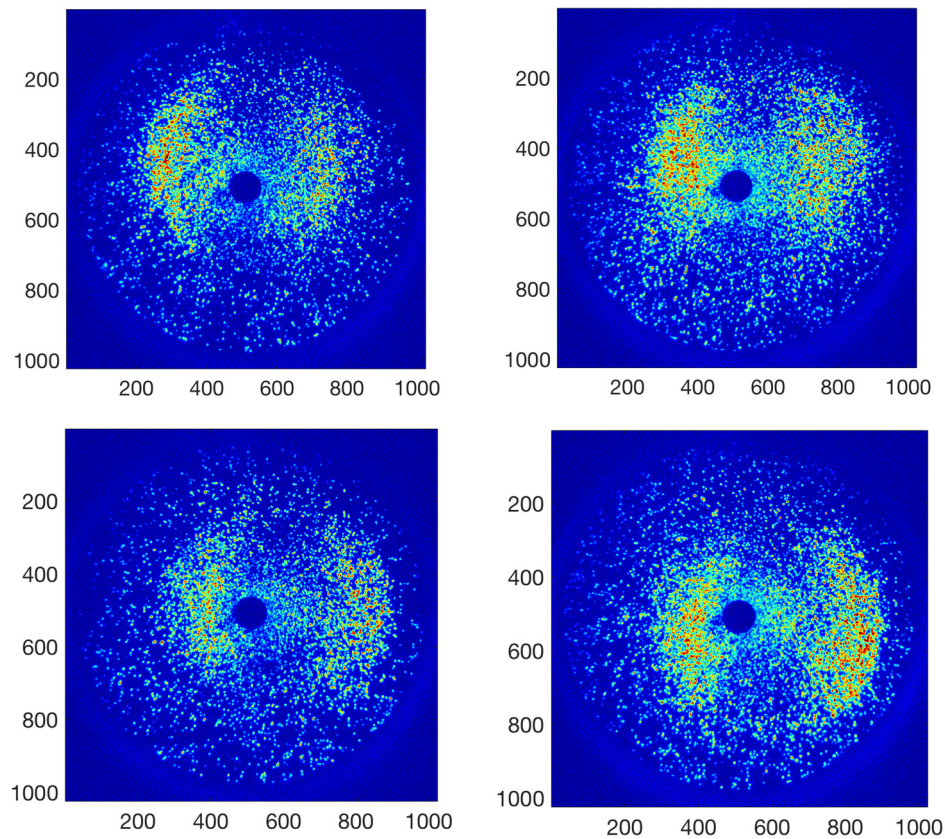
Attoclock:
Eckle, P., et al. "Attosecond ionization and tunneling delay time measurements in helium." *science* 322.5907 (2008): 1525-1529.
Eckle, Petrisa, et al. "Attosecond angular streaking." *Nature Physics* 4.7 (2008): 565.

XLEAP data

unstreaked

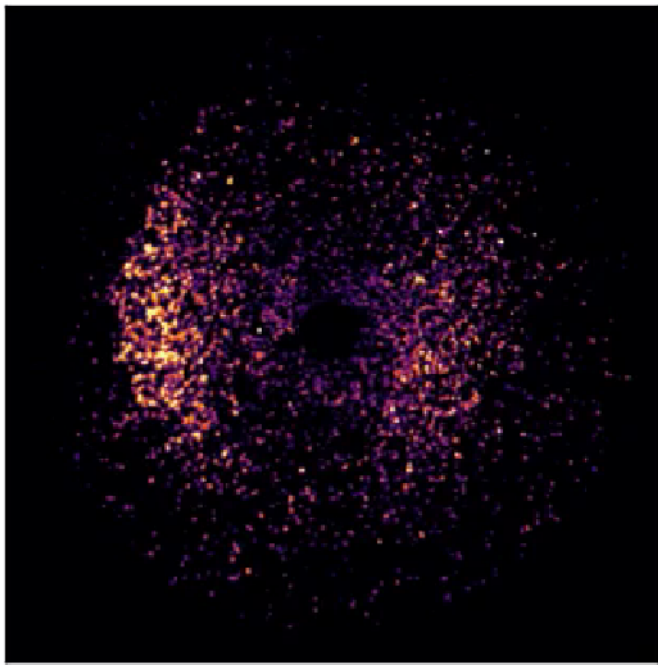


streaked

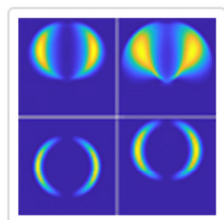
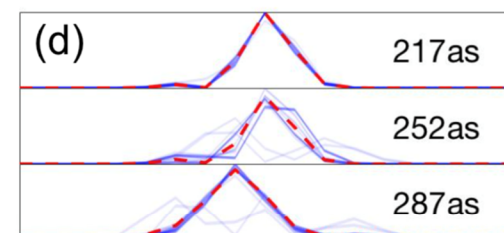
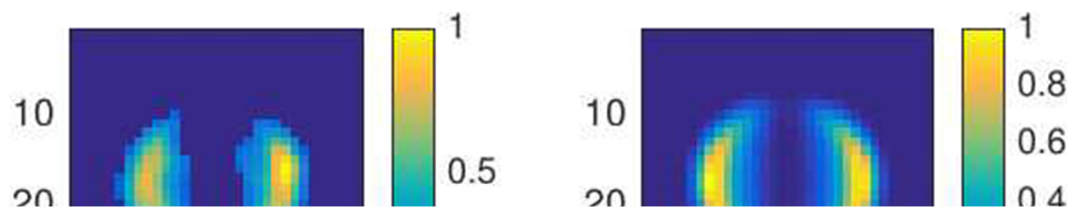


XLEAP data

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movie by Philipp Rosenberger



Characterizing isolated attosecond pulses with angular streaking

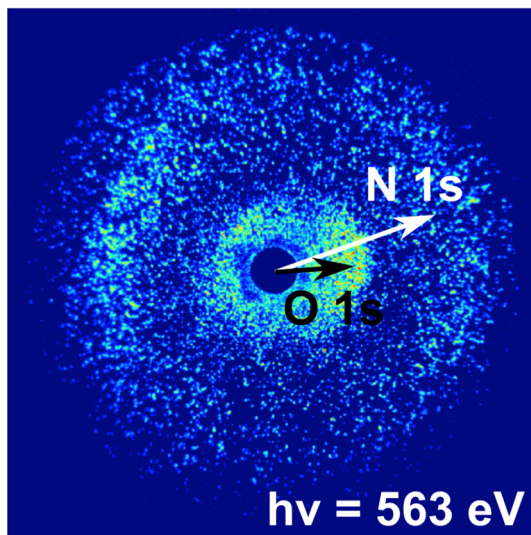
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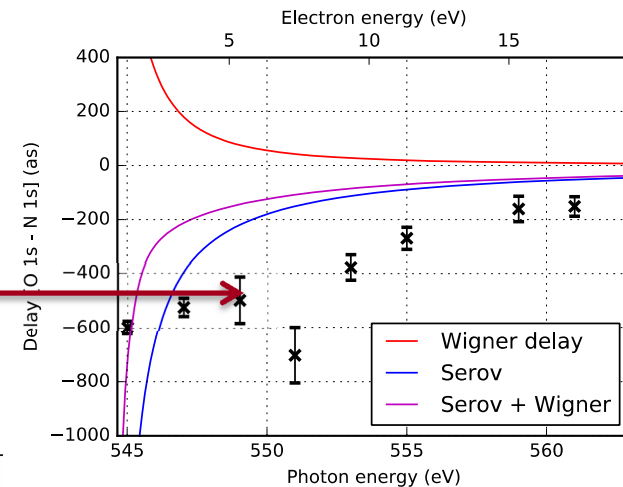
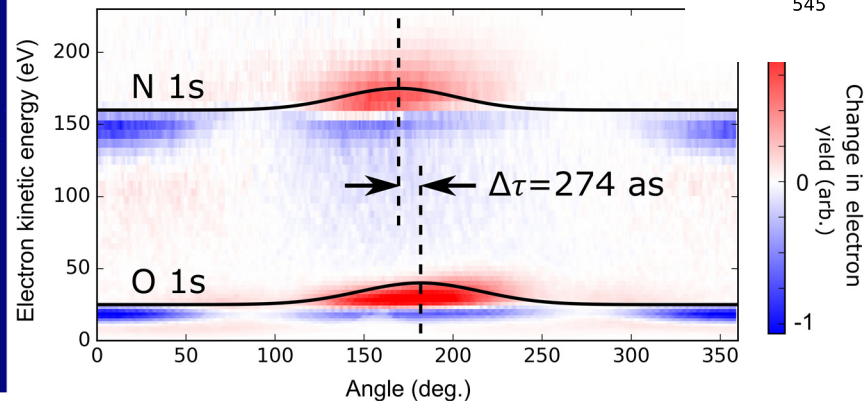
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Application: photoemission delay

Transient trapping of photoelectron wavepacket by the ionization potential leads to an increase in absorption cross section just above resonance.

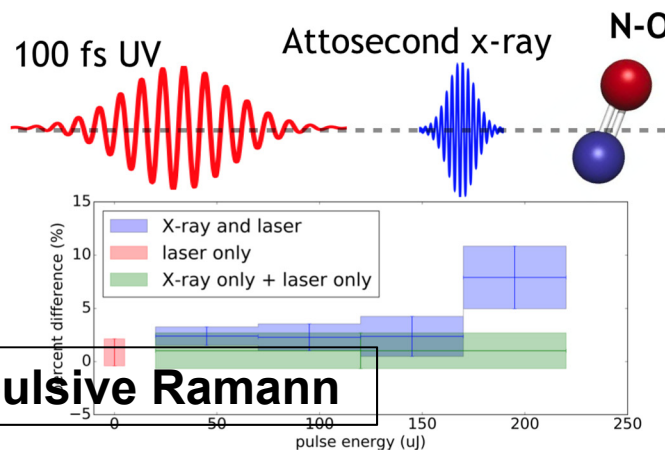
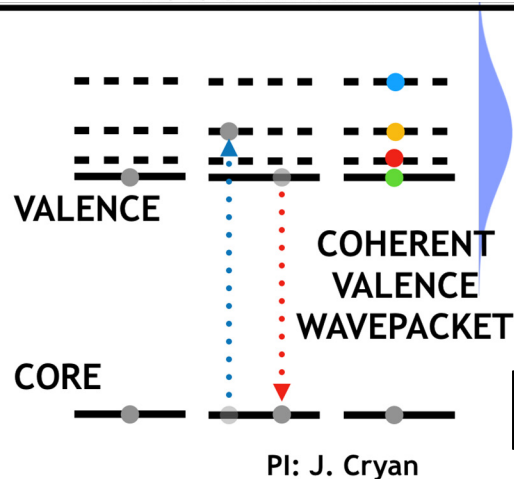
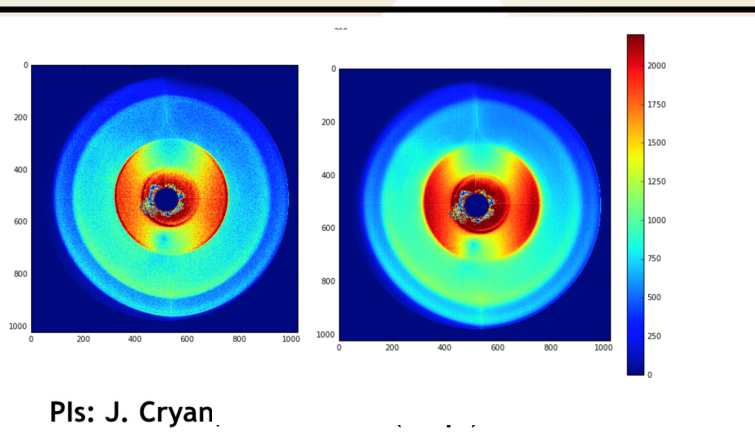
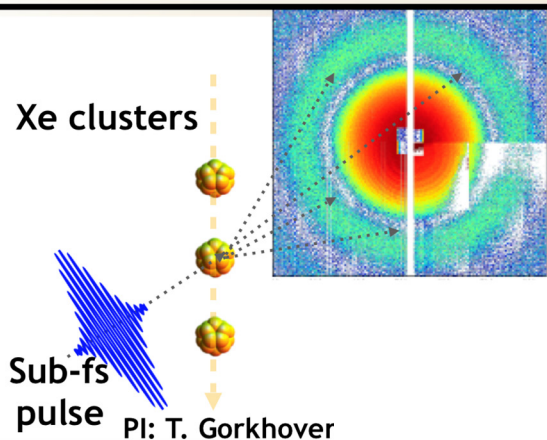


shape resonance



T. Driver

Attosecond science



Conclusions

- XLEAP: generating sub-fs x-ray pulses in FEL
- Angular streaking as a diagnostic tool for subfemtosecond x-ray pulses.
- Robust reconstruction algorithm that characterizes the time and frequency domains of the electron wavepacket.
- **XLEAP results: median of 280as, 480as for 905eV and 570eV photon energies, with 10s of uJ pulse energies.**
- Applications: Auger electrons, photoemission delay, and many other attosecond science studies.

Acknowledgements

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PULSE: J. Cryan, P. Bucksbaum, E. Champenois, T. Driver, A. Natan, J. O'Neal, R. Coffee, A. Gatton.

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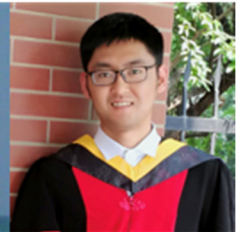
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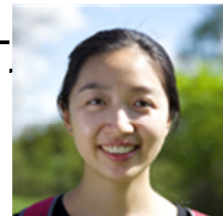


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Thank you! Questions?

